Suryansh Rohil

+91-9355284184 | sr738@snu.edu.in | LinkedIn | GitHub | Portfolio

Education

Shiv Nadar University — Noida, Delhi-NCR, India
B.Tech in Computer Science Engineering (6.27/10.0) — *Aug 2023 – Jul 2027*

GOVT SR SEC SCHOOL — Chamrori, Haryana
Class XII - HBSE (86.0%) — *Apr 2022 – Jul 2023*

D.A.V CENT PUBLIC SCHOOL — Raduar, Yamuna Nagar
Class X - CBSE (91.0%) — *Apr 2020 – Aug 2021*

Experience

Software Development Intern, *CodeClowns* — May 2025 – Aug 2025

- Architected a high-throughput recommendation engine for the **Dwij AI test-prep platform** using **Neo4j** and **Redis** edge caching.
- **Business Impact:** Reduced backend API call frequency by **15%** and accelerated study material retrieval times by **20%**, ensuring a scalable architecture for pilot users.

Backend Development Intern, *CodeClowns* — May 2024 – Oct 2024

- Streamlined 12+ **Go (Gin)** microservices using **Goroutines** and refactored **MongoDB/MariaDB** schemas to resolve state contention in high-concurrency environments.
- **Business Impact:** Achieved a **35% reduction in latency** and a **25% increase in data throughput** under concurrent load.

Projects

GradGuard: ML-Driven Threat Intelligence Honeypot

- Architected a high-interaction SSH honeypot in **Go**, dynamically provisioning isolated **Docker** containers and an egress network sinkhole with AWS honeytokens to securely capture and contain malware payloads.
- Engineered a dynamic deception engine and **eBPF** kernel monitor to actively mutate Linux filesystems against fingerprinting, driven by a real-time Machine Learning pipeline to classify attacker intent.
- Tech stack: Go, Docker, eBPF, Machine Learning, Linux.

GladMeds: AI-Powered Medical Assistant Platform

- Engineered a secure, full-stack medical assistant platform in **Next.js**, implementing a centralized Data Access Layer (DAL) to strictly govern the retrieval of sensitive health profiles and chronic conditions.
- Hardened application security by integrating Google SSO and AES-256 encryption for data at rest, protecting user medical data and real-time emergency SOS location tracking.
- Tech stack: Next.js, React, MongoDB.

GradLedger: A Decentralized Alumni–Student Mentorship Platform

- Architected a Zero Trust decentralized **Ethereum** platform, deploying immutable **Solidity** smart contracts to enforce cryptographic identity validation and role-based resource access.
- Engineered a defense-in-depth authentication perimeter integrating dual-factor verification and a Python biometric face-verification pipeline, backed by 10+ high-throughput **Go** REST APIs.
- Tech stack: Next.js, Solidity, Go, Python, MongoDB.

MorphDAG: A Workload-Aware Elastic DAG-Based Blockchain

- Orchestrated a highly concurrent execution engine for a DAG-based blockchain in **Go**, utilizing a hot account detection algorithm to mitigate state conflicts across parallel nodes.
- Fortified network consensus and accelerated transaction processing, achieving a verifiable **4.3x throughput speedup** and a **17.48% reduction** in global finality latency compared to standard EVM benchmarks.
- Tech stack: Go, Ethereum workload.

Technical Skills

Languages: Go (Golang), C/C++, Java, Solidity, Dart, JavaScript
Frameworks & Libraries: Flutter, Gin, Next.js, React, Nest.JS. Hyperledger Fabric
Databases & Caching: MongoDB, Neo4j, Redis, MySQL, MariaDB.
Tools & DevOps: Git, GCP, Docker, REST APIs

Achievements

Smart SNU Hackathon'25 Winner (1st place): Delivered a high-performance agricultural application, reducing memory overhead and accelerating rendering pipelines through efficient widget state management.